

11-14 Waves

Prerequisites

No prerequisites

The Main Points

- What is a wave?
 - A wave carries **energy** and can carry **information** without carrying **material** from one place to another.
 - Inside a wave, each part moves one way then the other. This is called an **oscillation**. Each part's oscillation causes the next part to oscillate.
- Longitudinal and transverse
 - In a **longitudinal** wave, the oscillations are parallel to the direction of energy transfer.
 - Sound is an example of a longitudinal wave.
 - In a **transverse** wave, the oscillations are at right angles to the direction of energy transfer.
- Parts of a wave
 - The high points on waves are called **peaks**.
 - The low points on waves are called **troughs**.
 - The midpoint (where things would be if there were no wave) is called the **equilibrium position**.
 - The height of the peaks above the midpoint is called the **amplitude**.
 - The depth of the troughs below the midpoint is also the **amplitude**.
 - In a longitudinal wave,
 - places where the particles have bunched together are called **compressions**.
 - places where the particles have spread out are called **rarefactions**.
 - the distance from one compression to the next (or one rarefaction to the next) is the wavelength.
- Repeating unit
 - If you take a 'freeze frame' picture of a wave, you can split it into identical repeating units.
 - Example: the section of wave from one peak to the next is a repeating unit.
 - The length of a repeating unit (the distance along the wave from one peak to the next, or one trough to the next) is called the **wavelength**, and has symbol λ (lambda).

Advanced Point (not necessary for KS3)

- Waves on the surface of water look transverse (the water level goes up and down as the wave itself moves sideways along the surface). However, a water wave actually involves water particles moving in circles, so it is partly transverse and partly longitudinal.

Teacher Quarter Briefing

- Introduction: <https://youtu.be/oEWdFINjkkU>
- Practice: https://isaacscience.org/question_decks#itsp_teach_wavesound
- Review: <https://youtu.be/hyF4JrTQ1cc>
- If you want to go further, please see the 'Sound and Waveform' materials

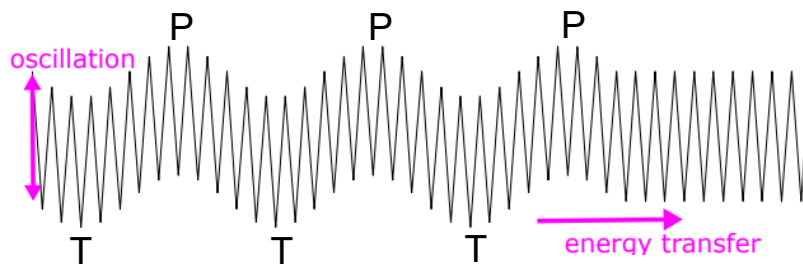
Class Question Notes

The worksheet can be printed either in full, or in cloze text form (where the red text is missing, and students can complete these blank spaces after class discussion). The online version of the notes requires the appropriate text to be dragged to the right place in the sentences.

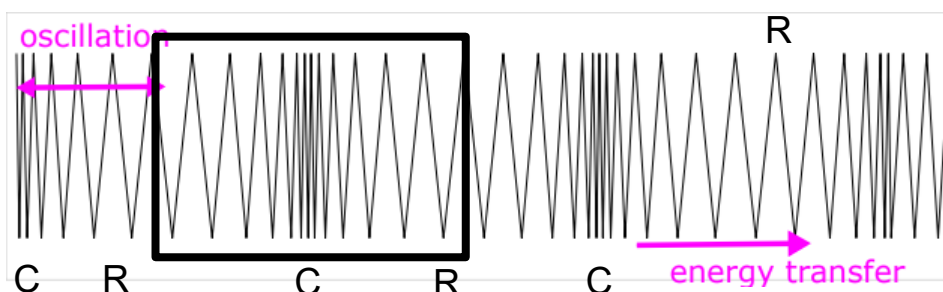
[Shallow learning gradient online assignment](#) - q1,2,4,6,7,8,9,10,11

[Steeper learning gradient online assignment](#) - q1,3,4,5,6,7,8,9,10,11

1. You may wish to go through this question with the class as a whole so that everything gets labelled well.
 - a. This is transverse because the oscillations are up-and-down while the energy moves to the right (oscillations are at right angles to the energy transfer direction).
 - b. See the diagram below for where the peaks and troughs are.



2. The distance from one peak to the next is about 3.7cm. Please accept answers in the range 3.5cm to 4.0cm.
3. The distance from one trough to the next is also about 3.7cm.
4. Students should notice that the two boxes look the same. This means that each box contains a repeating unit of the wave, and the length of the box will be the wavelength. Here each box contains the section of wave from one peak to the next. The box length is about 3.7cm.
5. Here the boxes also each contain one repeating unit, but here the repeating units are drawn from one trough to the next. The length of each box is still about 3.7cm, which is the wavelength.
6. This question draws together questions 2-5, that the repeating unit's length is the wavelength.
7. This is a longitudinal wave as the oscillations are left-and-right and the energy transfer direction is to the right. These are parallel. The compressions and rarefactions (together with a box giving one suggestion of a repeating unit) are shown below:



8. Here the students make another measurement of 3.7cm - this wave has the same wavelength as the earlier ones.
9. Peaks occur at distances of 0m, 5m, 10m, 15m. The spacing between these is 5m, so the wavelength is 5m.
10. In this question, students write down the amplitude and wavelength of four waves. You may need to remind students that the amplitude is the height of the peak above the midpoint. So in wave (a) the amplitude is 2cm (not 4cm, which is the height of the peak above the trough).
11. In this question, students complete boxes to describe how a citizen can listen to the news on the TV, and the part played by the speaker (which makes the sound wave), the air (through which the

sound wave travels) and the ear (which receives the sound wave). It starts 'The speaker in the TV oscillates.' You may wish to give this to the students to get them started.

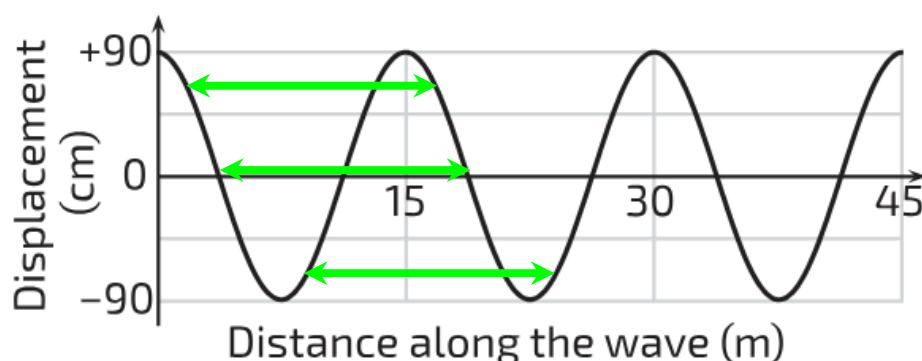
Homework Question Notes

These questions have a very similar form to the questions in the class task, so students can refer back to their earlier answers to help.

[Shallow learning gradient online assignment](#) - q1,2,3,4,5,6,7,8

[Steeper learning gradient online assignment](#) - q1,2,3,4,5,6,8,9,10

1. Students are given a situation and asked whether the wave is longitudinal or transverse. The rope is going up and down, but the friends are separated horizontally. The oscillation and the energy motion (from one friend to the other) are at right angles, so it is transverse. Students are also asked to label the peaks (high points) and troughs (low points).
2. Students are asked to mark one wavelength on a picture of a longitudinal wave. It is probably easiest for the student to draw this from one side of a compression to the same side of the next compression.
3. In this question, students use words from a list to explain how to measure a wavelength.
4. In this question students use words from a list to explain how to measure an amplitude.
5. In this question, students need to recognize that sound waves are always longitudinal. Therefore Groggy can be told that if he measures some transverse waves, then it must be an earthquake, whereas if he only measures longitudinal waves, then he can be sure it is Shirahoshi singing.
6. This question is similar to q10 on the class sheet, with students writing down the wavelength (λ) and amplitude of each of four waves.
7. Students are asked to pick the odd one out. Hint: can they identify whether each one is longitudinal or transverse? Only one is purely longitudinal (the sound wave).
8. Students choose whether each example is longitudinal or transverse. The first is sound, so must be longitudinal. The second (a ripple on a lake's surface) has the surface going up and down, whereas the energy travels out from the middle, so this looks transverse (actually it is both longitudinal and transverse as the water particles move in circles).
9. Students are asked to mark the wavelength three times on a wave. The key point here is it doesn't matter where the student starts, as long as they finish each label one repeating unit after it starts. An example of a correct answer is shown below.



10. We meet Shirahoshi and Groggy again as they attempt to communicate without using sound. Sound will cause the bottle to vibrate, but we won't see it moving back and forth.

Extension questions from 'Step Up to GCSE Physics':

https://isaacscience.org/question_decks#phys_step_up_35

https://isaacscience.org/question_decks#phys_step_up_36